

A Configurable Multimodal Interaction Toolkit for Voice and Gesture-Based 2D/3D Object Manipulation

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Abstract

We present a local-first multimodal interaction toolkit for prototyping voice- and gesture-based object manipulation in 2D and 3D applications. The toolkit separates modality adapters, temporal fusion, and application logic through a shared fusion core that converts speech and hand-gesture input into canonical application actions. We demonstrate the approach with a 3D shape-puzzle application in which speech specifies semantic intent, such as creating, selecting, rotating, and inserting objects, while gestures provide spatial input for pointing, dragging, palm-based rotation, and pinch-based resizing. To assess the toolkit, we compare the multimodal puzzle with a mouse-and-keyboard baseline that preserves the same task and scoring structure. The evaluation examines perceived usefulness, naturalness, precision, and developer-facing reusability in a questionnaire with 25 computer science students and a task-driven observation with 10 participants. Users perceived the toolkit as understandable and reusable, and rated the multimodal version positively for naturalness and usefulness in 3D manipulation, while the mouse-and-keyboard baseline remained stronger for precision, controllability, and completion time. Command-level observations further showed that the prototype was mostly functional, but that selection, speech recognition, and voice-gesture fusion were the main sources of failure. These findings suggest that voice and gesture can add perceived value for spatial manipulation tasks, but that their practical benefits depend on the trade-off between expressiveness, robustness, and control.

1 Introduction

Many interactive prototypes are still built around mouse and keyboard input, even when the task itself is spatial, embodied, or naturally distributed across multiple modalities. In this project, we investigate whether a reusable multimodal toolkit can make such interactions easier to prototype while keeping the application layer relatively simple.

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2026.

The concrete use case is a 3D shape puzzle. Users create and manipulate 3D objects to match target silhouettes. Instead of letting the application process raw microphone or camera data directly, the system routes voice and gesture events through a shared fusion core that produces canonical actions for the app. This separation is the central contribution of the project: the application does not need to know how speech recognition or hand tracking works internally, only how to react to actions such as `create_object`, `set_interaction_mode`, `move_object`, or `insert_object`, or others, depending on the configuration that a person made for their specific use case.

Our research hypothesis is that multimodal interaction adds perceived value for 2D/3D object manipulation because speech is well-suited for semantic intent and gesture is well-suited for spatial reference [2, 4, 6]. At the same time, we do not expect multimodal input to dominate conventional input in every dimension, but rather to create a trade-off where the conventional approach has its own benefits, such as precision, reliability, and possibly faster task completion [1, 3], while the multimodal interface can bring greater naturalness and ease of use [5, 6].

The source code and setup instructions for the toolkit, the 3D shape puzzle application, and the mouse/keyboard baseline are available in an accompanying GitHub repository.⁰

2 Toolkit Architecture

The project is implemented as a reusable multimodal interaction toolkit with two applications built on top of the same core: the 3D shape puzzle and a 2D/3D sandbox demo. The goal of these two apps is not just to build them, but to make sure input-processing logic is separated from the application logic, so that the same voice, gesture, and fusion components can be reused in other interaction prototypes, as shown in Figure 1.

2.1 System Structure

The architecture follows a local-first design. Speech recognition runs through a voice module, while hand tracking and gesture recognition run through a webcam/gesture module. These modules

⁰https://github.com/bohdanpotuzhnyi/JMCS_MMI_course_project/

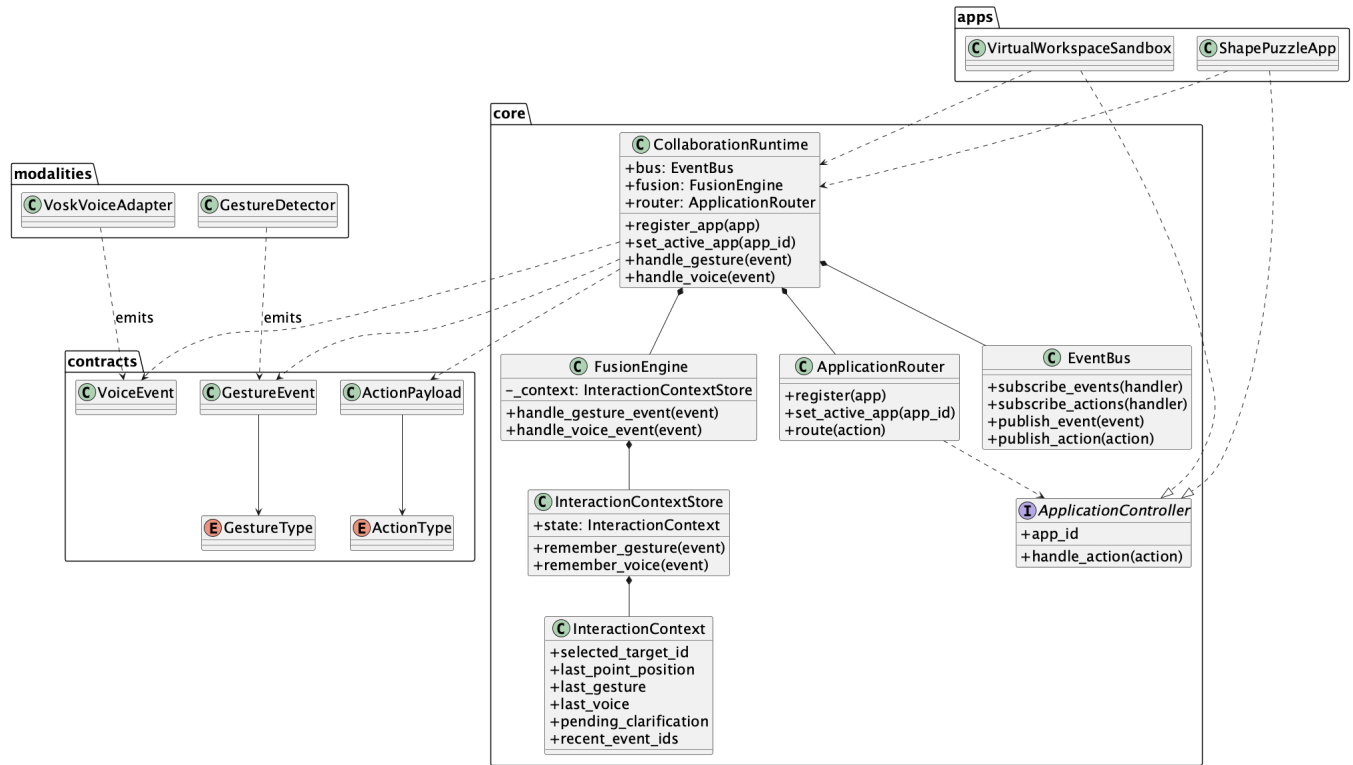


Figure 1: Implemented system

are responsible for processing raw input and emitting normalized modality events.

The voice path is implemented by `VoskVoiceAdapter`. It uses the `SpeechRecognition` microphone wrapper together with `PyAudio` for capture, but performs speech-to-text locally with a Vosk model instead of a remote API. In the current repository setup, the local model is `vosk-model-en-us-0.22-lgraph`. Recognition is constrained by a command grammar built from the configured phrase lists, and the recognized transcript is normalized and mapped to an intent in `intent_from_transcript.py` through substring rules with lightweight fuzzy recovery for common speech-to-text errors. The output is a `VoiceEvent` carrying an event id, timestamp, confidence, transcript, `is_final` flag, and an optional normalized intent label such as `create-cube`, `drag`, or `insert`.

The gesture path is implemented by `GestureDetector`. It uses `OpenCV` for camera capture and `MediaPipe Hand Landmarker` for local hand tracking, producing up to 21 normalized hand landmarks per detected hand. In the current repository setup, the local hand-tracking asset is `hand_landmarker.task`. Gesture classification is rule-based: static classifiers recognize `point`, `pinch`, `grab`, `fist`, and `open_palm`, while a short rolling history detects directional swipes. Spatial features are derived from the palm center, the index-finger tip, and the thumb-index pinch distance, which are later reused by the fusion layer for pointing, dragging, rotation, and resizing. The module emits a `GestureEvent` with an event id, timestamp, confidence, normalized position, optional full landmark list, and handedness.

The modality events are passed to `CollaborationRuntime` and `FusionEngine`, which combine recent voice and gesture evidence into canonical `ActionPayload` objects. Applications consume these payloads and update their own scene state, rendering, and task logic.

This structure keeps modality logic, fusion logic, and application logic independent. It also makes the interaction behavior easier to inspect and configure, because applications react to action-level events instead of raw microphone or camera input.

2.2 Canonical Contracts

The toolkit standardizes communication through contracts such as `GestureEvent`, `VoiceEvent`, and `ActionPayload`. This is important for reuse because applications react to a stable vocabulary of actions instead of raw input streams.

In the current implementation, `ActionPayload` can carry a canonical action type together with optional position, delta, scale, rotation, `object_type`, mode, metadata, and `source_events`. A `target_id` field also exists in the shared contract, although it is not the main mechanism used by the shape-puzzle flow.

For example, the application does not need to know whether a target position came from a pointing gesture, a drag gesture, or a stored interaction context. It only receives action-level commands such as `create_object`, `select_object`, `move_object`, and `update_pointer`. The current shape-puzzle configuration also uses `set_interaction_mode`, `insert_object`, `delete_object`, and `reset_app`. Rotation and resizing are realized through `set_interaction`

_mode together with continuous update_pointer events rather than through dedicated rotate_object or resize_object actions.

2.3 Fusion Strategy

The current implementation uses a rule-based fusion engine. Voice contributes semantic intent, while gesture contributes spatial evidence and manipulation state. In the shape-puzzle configuration, speech specifies intents such as create-*, select, move-here, drag, rotate, resize, done, insert, restart, and delete. Gesture contributes the current pointer position, selection evidence from point/pinch/grab, palm-motion deltas, and pinch-distance changes. This means that object rotation and resizing are not triggered by standalone gesture labels alone; instead, the voice command sets an interaction mode, and subsequent pointer updates provide the continuous palm or pinch signal used by the application.

The runtime also keeps a short-lived interaction context so that a spoken command can reuse the most recent gesture-derived position or selected object. Locally, this state is maintained by InteractionContextStore, which remembers the last gesture and voice event, the last point-derived position, and a bounded list of recent event ids. In the current 3D configuration, voice and gesture are fused when they fall within a temporal window of 1.5s; the context store keeps up to 20 recent event ids, but the main practical reuse mechanism is this short voice-gesture window, together with the remembered pointer position.

2.4 Configuration

The repository uses configuration files to map spoken phrases to intents and intents to actions. This reduces the amount of application-specific parsing code and makes it possible to adapt the vocabulary to new tasks without rewriting the core architecture. For the shape-puzzle app, the main file is src/apps/shape-puzzle/fusion-3d.conf. It uses an INI-style structure with separate sections for core behavior, timing, confidence thresholds, pointer emission, voice phrase inventories, and action rules. In practice, this lets a developer change command phrases, enable or disable intents or gestures, adjust the fusion window, and remap the same input vocabulary to different action payloads without changing the application code.

```
[fusion-core]
ENABLED = YES
ENGINE = 3d
STRATEGY = rule-based
```

```
[fusion-timing]
TEMPORAL_WINDOW = 1.5s
```

```
[fusion-confidence]
VOICE_MIN_CONFIDENCE = 0.60
GESTURE_MIN_CONFIDENCE = 0.60
FUSED_MIN_CONFIDENCE = 0.65
```

```
[fusion-pointer]
SOURCE = index-finger
EMIT_UPDATE_POINTER = YES
EMIT_PALM_DELTA = YES
EMIT_PINCH_DELTA = YES
```

```
[fusion-voice-include]
INTENTS = create-cube, select, ...
```

```
[fusion-voice.intent.create-cube]
PHRASES = create cube, make cube, add cube
```

```
[fusion-action.voice.create-cube]
ACTION = create_object
OBJECT_TYPE = cube
USE_POSITION = YES
```

3 Applications

3.1 3D Shape Puzzle

The main application in this paper is a 3D shape puzzle implemented in src/apps/shape-puzzle/app.py. The user creates different shapes such as cubes, cuboids, spheres, and diamonds, then moves, rotates, and resizes them to match target silhouettes. The app uses custom 3D math and painter-style depth sorting. This application is great for a demonstration task, as responsibility is divided across modalities. As such, speech contributes semantic intent, including commands for object creation, selection, rotation mode, resize mode, and insertion, while gesture contributes spatial reference and continuous manipulation information such as pointer position, dragging, palm-motion updates, and pinch-based scaling. It is showcased in Figure 2.

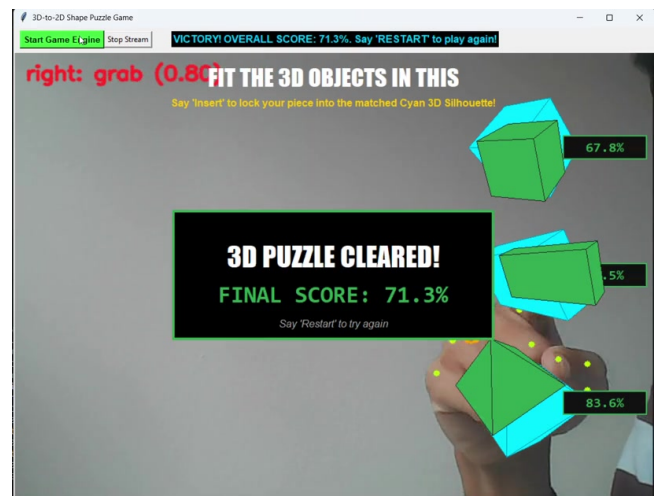


Figure 2: Multimodal 3D shape puzzle

3.2 Virtual Workspace Sandbox

The repository also contains a sandbox-style application, which was the first app that we made after the creation of the fusion core to make sure it actually works, and later we kept using it for testing the different features that we were adding to the core, as well as making sure that the 3D puzzle app is independent of the core and is not breaking it. One can find it in the src/apps/virtual-workspace/app.py. The app is pictured in Figure 3.

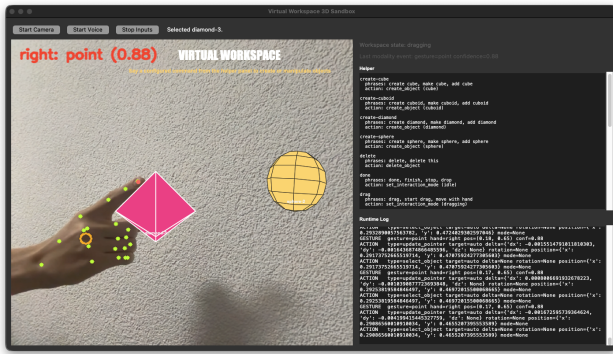


Figure 3: Virtual workspace sandbox

3.3 Mouse/Keyboard Baseline

Mainly for evaluation reasons, the repository also contains what we call a "baseline" version in `src/apps/shape-puzzle/mouse_keyboard_app.py`. It preserves the same game goal, target rendering, and scoring logic as "3D Shape Puzzle", but replaces multimodal input with a "conventional" desktop UI based on mouse selection, drag operations, toolbar controls, side-panel actions, and keyboard shortcuts. It is showcased in Figure 4.

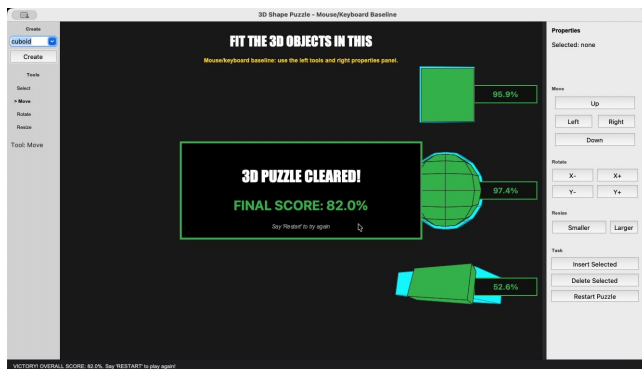


Figure 4: Mouse/keyboard baseline for the 3D puzzle

4 Evaluation Design

We use two evaluation instruments. First, a post-demo questionnaire was completed by 25 computer science students. This questionnaire evaluates the perceived clarity, reusability, and usefulness of the toolkit, as well as the perceived trade-off between the multimodal puzzle and the mouse/keyboard baseline. Second, a task-driven observation was conducted with 10 participants. During this observation, we recorded command-level success and failure for a predefined sequence of multimodal actions. We also recorded task completion times for the multimodal puzzle and the mouse/keyboard baseline.

4.1 Research Questions

The evaluation is organized around two research questions:

- (1) Does the toolkit appear understandable and reusable for building other voice- and gesture-based interaction prototypes?
- (2) Does the multimodal interaction style bring perceived value for the presented 3D object manipulation task compared with the mouse/keyboard baseline?

The first question focuses on the toolkit as a software design. It is evaluated through questionnaire items about whether the goal of the toolkit was easy to understand, whether the toolkit seemed reusable beyond the shown demo, and whether participants could imagine it being useful for prototyping other interactive applications.

The second question focuses on the interaction experience. It is evaluated through questionnaire items comparing the multimodal version with the mouse/keyboard baseline. These items cover whether the multimodal version felt more natural for object manipulation, whether it looked easier for rotating and resizing 3D objects, and whether the mouse/keyboard baseline looked more precise and controllable.

The central hypotheses evaluated in this project are:

- **H0:** multimodality does not add perceived value compared with the mouse/keyboard baseline.
- **H1:** multimodality adds perceived value for the presented 3D manipulation task.

In this evaluation, perceived value is measured through categories such as toolkit understandability, perceived reuse potential, usefulness of combining voice and gesture, naturalness for object manipulation, ease of rotation, and resizing, perceived precision and controllability, and overall preference.

4.2 Questionnaire Design

The questionnaire contains eight items and was answered by 25 users. The first seven use a five-point Likert scale from strongly disagree to strongly agree. The final item asks for an overall preference between the mouse/keyboard baseline, the multimodal version, or no clear preference.

The questionnaire is split into three blocks. The first block targets toolkit-level perception: clarity of the toolkit goal, perceived usefulness of combining voice and gesture, perceived reusability beyond the demo, and usefulness for prototyping other applications. These items address the first research question.

The second block targets application-level comparison. Participants rated whether the multimodal version felt more natural for object manipulation, whether it looked easier for rotating and resizing 3D objects, and whether the mouse/keyboard baseline looked more precise and controllable.

The final third block is a preference question whose purpose is to capture a summary judgment for the specific puzzle task, while still allowing participants to choose no clear preference.

4.3 Task-Driven Observation

In addition to the questionnaire, we conducted a task-driven observation with 10 participants. During each run, participants carried out a predefined sequence of multimodal actions in the shape puzzle.

The observer-coded table records the requested speech command, the expected gesture or action, the visible system response, whether the command worked correctly, a numeric correctness score, and a coarse error label. We use this table to estimate which parts of the multimodal pipeline were reliable and which parts caused errors.

The same observation also collected completion times for the multimodal puzzle and the mouse/keyboard baseline. Ten participants had complete paired timing data and are included in the paired time analysis. Since the sample is small and all observed differences point in the same direction, we use an exact paired sign test rather than a parametric comparison.

5 Results

The questionnaire contains 25 responses. The first block consists of four toolkit-oriented questions with the following results:

- "The goal of the toolkit was easy to understand" mean of **4.24**, SD **0.6**
- "Combining voice and gesture seemed useful for object manipulation" mean of **3.96**, SD **0.73**
- "The toolkit seemed reusable beyond the shown demo app" mean of **3.92**, SD **0.81**
- "I could imagine this toolkit being useful for prototyping other interactive applications" mean of **3.96**, SD **0.54**

For all four toolkit-oriented items, exact sign tests against the neutral midpoint showed that responses were significantly above neutral ($p < .001$). This allows us to state that participants in this questionnaire understood the goal of the toolkit and perceived it as useful and reusable outside the scope of the presented demo.

The second block of questions was a bit more mixed but still useful, as such:

- "Compared with the mouse/keyboard baseline, the multimodal version felt more natural for object manipulation" mean of **3.68**, SD **0.85**
- "Compared with the mouse/keyboard baseline, the multimodal version looked easier for rotating and resizing 3D objects" mean of **4.00**, SD **0.96**
- "Compared with the multimodal version, the mouse/keyboard baseline looked more precise and controllable" mean of **4.12**, SD **0.67**

Running a sign test against the neutral midpoint resulted in the following results. Participants rated the multimodal version as more natural than the mouse/keyboard baseline ($p < .01$) and easier for rotating and resizing 3D objects ($p < .001$). At the same time, participants also rated the mouse/keyboard baseline as more precise and controllable ($p < .001$). From this, we can state that, as we predicted, the multimodal toolkit is not going to be the best in all aspects, but rather a trade-off: it provides the user with a feeling of naturalness and easier spatial control, while the mouse gained an advantage in precision and control.

The third block and final question, "Overall, for this specific puzzle task I would prefer:" resulted in 12 participants preferred the multimodal version and 13 reported no clear preference. No participant preferred the mouse/keyboard baseline for this task. Running any test, such as a two-sided exact binomial test, would result in $p < .001$.

For the task-driven observation, we treat each requested command as one positive command attempt. Since the observation table only records requested actions, it supports true-positive and false-negative style analysis. Across 10 participants, the observation contains 85 logged multimodal command attempts. Out of these, 70 commands were successful, 4 were partially successful, and 11 failed. This gives a full success rate of $70/85 = 82.4\%$, a partial-success rate of $4/85 = 4.7\%$, and a failure rate of $11/85 = 12.9\%$. Using the weighted correctness score, the overall mean score is 0.847.

Table 1: Task-driven observation results by requested action.

Action	Count	True pos.	Partial	Failed	Score
create_cube	10	10	0	0	1.00
create_sphere	10	10	0	0	1.00
select	30	21	0	9	0.70
drag	10	6	4	0	0.80
rotate	10	8	0	2	0.80
move_here	5	5	0	0	1.00
resize	5	5	0	0	1.00
done	5	5	0	0	1.00
Total	85	70	4	11	0.85

As one can see from the table 1, certain actions had better and worse scores. As such, the select command had a mean score of 0.70 across 30 attempts, with 9 failures. The drag command had a mean score of 0.80, with four partial failures caused by incomplete fusion of speech and gesture. The rotate command also had a mean score of 0.80, with two failures. Error labels in the table show 10 recognition errors and 6 fusion-related errors, so recognition appears to be the dominant immediate source of failure, but fusion reliability is also an identifiable issue.

A final evaluation metric that we got from this task-driven study is completion time. As such, for 10 participants, the multimodal condition had a mean completion time of 102.6 seconds, compared with 62.2 seconds for the mouse/keyboard baseline. Given the small sample and the clear direction of all differences, the most appropriate test here is an exact paired sign test rather than a parametric t-test. The resulting one-sided p -value is 0.00098, indicating that the baseline (mouse/keyboard) was reliably faster in this sample.

Having this, we can summarize the results for the hypotheses stated earlier. We can reject **H0** and support **H1** for the following dimensions:

- **Naturalness:** participants rated the multimodal version as more natural for object manipulation than the mouse/keyboard baseline.
- **Rotation and resizing:** participants rated the multimodal version as easier for rotating and resizing 3D objects.
- **Overall clear preference:** among participants who expressed a clear preference, all preferred the multimodal version over the mouse/keyboard baseline.

At the same time, we cannot reject **H0** for all dimensions. The results support **H0**, or at least fail to support **H1**, for the following dimensions:

- **Precision and controllability:** participants rated the mouse/keyboard baseline as more precise and controllable than the multimodal version. Seeing errors in task processing, we can also say that this is confirmed not only by the p-value from the questionnaire, but also by failed user actions.
- **Task completion time:** the mouse/keyboard baseline was significantly faster than the multimodal version in the paired timing data.

The final result is, therefore, not that multimodality replaces conventional input, but that it offers a useful trade-off for spatial interaction when the recognition and fusion pipeline is robust enough.

6 Discussion

Overall, the project was not fully successful in every dimension, but it was also clearly not a failure. We implemented a working multimodal toolkit, built two applications on top of the same core, and evaluated both the idea of the toolkit and the behavior of the current prototype. The results show that the architecture makes sense to participants and that multimodal interaction can bring perceived value, but also that the current implementation still has clear practical limitations.

The main finding is that multimodal interaction is not the answer to all interaction problems, at least not in the state in which we implemented it. The evaluation results suggest that the hypothesis is only partially supported. Multimodality did bring value in terms of naturalness, spatial manipulation, and user interest. However, it did not bring value in terms of speed, precision, and robustness.

With the help of the task-driven observation, we gained an understanding of where the prototype needs improvement. Most commands worked, but failures were concentrated around selection, speech recognition, and fusion between voice and gesture.

6.1 Limitations

There are several limitations in our evaluation. First, the questionnaire was answered by computer science students. This group is useful for evaluating the idea of a reusable toolkit, because they can understand software architecture and prototyping concepts. However, it also means that the evaluation is biased toward technical users. The results may look different with non-technical users or with users from a specific application domain.

Second, the questionnaire was anonymous. This made it easier for participants to answer freely, but it also means that we cannot strictly verify that nobody answered twice. We asked participants to answer honestly, and we believe this was followed, but it is still a limitation of the collected data.

Third, not all questionnaire participants used the system directly. The questionnaire was based on the toolkit explanation and demonstrations of the multimodal puzzle and the mouse/keyboard baseline. This is enough to evaluate perception, understandability, and first impressions, but it is weaker than a controlled hands-on study where every participant performs the same task with both interfaces.

Finally, the task-driven observation was small and partly manual. It involved 10 participants. Some measurements, especially completion time and command outcome labels, were recorded during human observation. Therefore, they may be less precise than data collected automatically by the system. The results are still useful for identifying trends and failure points, but they should not be treated as a final benchmark.

6.2 Future Work

The next step is to improve the weak parts identified during the evaluation. The most important improvements are better speech recognition, more reliable selection, clearer feedback when the system does not understand the user, and more robust fusion between voice and gesture. After these improvements, the toolkit should be evaluated again in a more controlled offline study with more participants from different domains and direct hands-on use of both interfaces.

Another direction is to test the toolkit in more specific use cases (KYC procedures). Some informal discussions showed interest in using this kind of voice-and-gesture interaction for domain-specific workflows, where users need to inspect or manipulate visual objects.

In summary, the project showed both the potential and the limits of our multimodal toolkit. It confirmed that voice and gesture can make some 3D manipulation tasks feel more natural, but it also showed that practical value depends heavily on robustness, precision, and feedback.

7 Author Contributions

- **Voice module:** Assaf Cohen
- **Gesture module:** Sai Sharan Karanam
- **Fusion core:** Bohdan Potuzhnyi
- **3D puzzle game:** Vamshi Krishna Neelam
- **Documentation:** Vlada Svirsh
- **Evaluation, review, presentation and editing:** all authors

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A Appendix

Table 2: Completion times for the multimodal and mouse/keyboard conditions.

Participant	Multimodal time (s)	Mouse/keyboard time (s)
P1	72	50
P2	172	96
P3	189	79
P4	81	45
P5	62	48
P6	88	43
P7	137	65
P8	95	78
P9	59	52
P10	71	66
Mean	102.6	62.2

Table 3: Questionnaire students' responses.

ID	The goal of the toolkit was easy to understand.	Combining voice and gesture seemed useful for object manipulation.	The toolkit seemed reusable beyond the shown demo app.	I could imagine this toolkit being useful for prototyping other interactive applications	Compared with the mouse/keyboard baseline, the multimodal version felt more natural for object manipulation.	Compared with the mouse/keyboard baseline, the multimodal version looked easier for rotating and resizing 3D objects.	Compared with the multimodal version, the mouse/keyboard baseline looked more precise and controllable.	Overall, for this specific puzzle task I would prefer:
1	Agree	Agree	Strongly agree	Strongly agree	Agree	Strongly disagree	Agree	no clear preference
2	Agree	Agree	Strongly agree	Agree	Agree	Strongly agree	Neutral	multimodal version
3	Agree	Strongly agree	Disagree	Agree	Agree	Strongly agree	Agree	no clear preference
4	Strongly agree	Agree	Agree	Strongly agree	Agree	Strongly agree	Agree	multimodal version
5	Agree	Agree	Neutral	Agree	Agree	Agree	Strongly agree	no clear preference
6	Agree	Strongly agree	Agree	Agree	Neutral	Agree	Agree	no clear preference
7	Strongly agree	Agree	Agree	Agree	Agree	Agree	Agree	multimodal version
8	Agree	Agree	Agree	Neutral	Agree	Neutral	Strongly agree	multimodal version
9	Agree	Neutral	Agree	Agree	Disagree	Agree	Agree	no clear preference
10	Neutral	Agree	Agree	Agree	Neutral	Agree	Neutral	multimodal version
11	Strongly agree	Strongly agree	Agree	Agree	Agree	Strongly agree	Agree	multimodal version
12	Agree	Agree	Disagree	Agree	Neutral	Agree	Agree	no clear preference
13	Strongly agree	Agree	Strongly agree	Agree	Agree	Strongly agree	Neutral	multimodal version
14	Agree	Agree	Agree	Agree	Strongly agree	Agree	Agree	no clear preference
15	Agree	Disagree	Agree	Neutral	Neutral	Agree	Strongly agree	no clear preference
16	Strongly agree	Agree	Agree	Agree	Agree	Strongly agree	Neutral	multimodal version
17	Agree	Strongly agree	Agree	Agree	Strongly agree	Agree	Strongly agree	multimodal version
18	Strongly agree	Agree	Strongly agree	Strongly agree	Agree	Agree	Agree	no clear preference
19	Agree	Agree	Agree	Neutral	Agree	Disagree	Agree	multimodal version
20	Agree	Neutral	Agree	Agree	Disagree	Agree	Strongly agree	no clear preference
21	Agree	Neutral	Agree	Agree	Disagree	Agree	Strongly agree	no clear preference
22	Strongly agree	Strongly agree	Strongly agree	Agree	Agree	Strongly agree	Agree	multimodal version
23	Neutral	Agree	Neutral	Agree	Neutral	Agree	Agree	no clear preference
24	Agree	Neutral	Agree	Neutral	Agree	Neutral	Agree	no clear preference
25	Strongly agree	Agree	Neutral	Agree	Strongly agree	Agree	Strongly agree	multimodal version